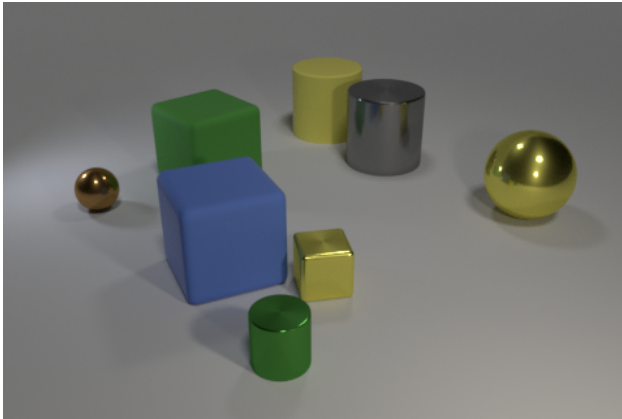




What color is the cylinder in front of the yellow cube?

```
1 {'directions': {'above': [0.000, 0.000, 1.000],
2                     'behind': [-0.754, 0.656, 0.000],
3                     'below': [-0.000, -0.000, -1.000],
4                     'front': [0.754, -0.656, -0.000],
5                     'left': [-0.656, -0.754, 0.000],
6                     'right': [0.656, 0.754, -0.000]},
7   'image_filename': 'CLEVR_val_011568.png',
8   'image_index': 11568,
9   'objects': [{'3d_coords': [-0.551, 2.301, 0.700],
10                  'color': 'gray',
11                  'material': 'metal',
12                  'pixel_coords': [293, 96, 12.618],
13                  'rotation': 187.432,
14                  'shape': 'cylinder',
15                  'size': 'large'},
16                {'3d_coords': [0.430, -2.164, 0.700],
17                  'color': 'blue',
18                  'material': 'rubber',
19                  'pixel_coords': [168, 173, 9.256],
20                  'rotation': 46.785,
21                  'shape': 'cube',
22                  'size': 'large'},
23                {'3d_coords': [-2.185, -0.509, 0.700],
24                  'color': 'green',
25                  'material': 'rubber',
26                  'pixel_coords': [157, 108, 11.937],
27                  'rotation': 282.699,
28                  'shape': 'cube',
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32                  'material': 'rubber',
33                  'pixel_coords': [249, 75, 13.944],
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```

34         'rotation': 55.072,
35         'shape': 'cylinder',
36         'size': 'large'},
37     {'3d_coords': [2.420, 2.343, 0.700],
38         'color': 'yellow',
39         'material': 'metal',
40         'pixel_coords': [404, 133, 10.751],
41         'rotation': 273.867,
42         'shape': 'sphere',
43         'size': 'large'},
44     {'3d_coords': [2.408, -2.868, 0.350],
45         'color': 'green',
46         'material': 'metal',
47         'pixel_coords': [213, 257, 7.728],
48         'rotation': 233.806,
49         'shape': 'cylinder',
50         'size': 'small'},
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52         'color': 'brown',
53         'material': 'metal',
54         'pixel_coords': [72, 143, 11.096],
55         'rotation': 70.238,
56         'shape': 'sphere',
57         'size': 'small'},
58     {'3d_coords': [1.649, -1.450, 0.350],
59         'color': 'yellow',
60         'material': 'metal',
61         'pixel_coords': [246, 201, 9.081],
62         'rotation': 140.802,
63         'shape': 'cube',
64         'size': 'small'}]],
65 'relationships': {'behind': [[3],
66                             [0, 2, 3, 4, 6],
67                             [0, 3],
68                             [],
69                             [0, 2, 3, 6],
70                             [0, 1, 2, 3, 4, 6, 7],
71                             [0, 2, 3],
72                             [0, 1, 2, 3, 4, 6]],
73         'front': [[1, 2, 4, 5, 6, 7],
74                  [5, 7],
75                  [1, 4, 5, 6, 7],
76                  [0, 1, 2, 4, 5, 6, 7],
77                  [1, 5, 7],
78                  [],
79                  [1, 4, 5, 7],
80                  [5]],
81         'left': [[1, 2, 3, 5, 6, 7],

```

```

82         [2, 6],
83         [6],
84         [1, 2, 5, 6, 7],
85         [0, 1, 2, 3, 5, 6, 7],
86         [1, 2, 6],
87         [],
88         [1, 2, 5, 6]],
89     'right': [[4],
90               [0, 3, 4, 5, 7],
91               [0, 1, 3, 4, 5, 7],
92               [0, 4],
93               [],
94               [0, 3, 4, 7],
95               [0, 1, 2, 3, 4, 5, 7],
96               [0, 3, 4]]},
97     'split': 'val'}

```

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1  {'answer': 'green',
2   'image_filename': 'CLEVR_val_011568.png',
3   'image_index': 11568,
4   'program': [{ 'function': 'scene', 'inputs': [], 'value_inputs': []},
5                { 'function': 'filter_color',
6                  'inputs': [0],
7                  'value_inputs': ['yellow']}],
8                { 'function': 'filter_shape',
9                  'inputs': [1],
10                 'value_inputs': ['cube']}],
11                { 'function': 'unique', 'inputs': [2], 'value_inputs': []},
12                { 'function': 'relate', 'inputs': [3], 'value_inputs': ['front',
13                               '']}],
14                { 'function': 'filter_shape',
15                  'inputs': [4],
16                  'value_inputs': ['cylinder']}],
17                { 'function': 'unique', 'inputs': [5], 'value_inputs': []},
18                { 'function': 'query_color', 'inputs': [6], 'value_inputs': []}
19                ]],
18  'question': 'What color is the cylinder in front of the yellow cube?',
19  'question_family_index': 75,
20  'question_index': 115673,
21  'split': 'val'}

```

Record, q_{rec} associated with question:

$$\left[\begin{array}{l} \text{bg} = \left[\begin{array}{l} x : Ind \\ x_{\text{color}} : \langle \lambda v : Ind . \text{yellow}(v), \langle x \rangle \rangle \\ x_{\text{shape}} : \langle \lambda v : Ind . \text{cube}(v), \langle x \rangle \rangle \\ y : Ind \\ yx_{\text{front}} : \langle \lambda u : Ind . \lambda v : Ind . \text{front}(u, v), \langle y, x \rangle \rangle \\ y_{\text{shape}} : \langle \lambda v : Ind . \text{cylinder}(v), \langle y \rangle \rangle \end{array} \right] \\ \text{clfr} = \lambda r : \left[\begin{array}{l} x : Ind \\ x_{\text{color}} : \langle \lambda v : Ind . \text{yellow}(v), \langle x \rangle \rangle \\ x_{\text{shape}} : \langle \lambda v : Ind . \text{cube}(v), \langle x \rangle \rangle \\ y : Ind \\ yx_{\text{front}} : \langle \lambda u : Ind . \lambda v : Ind . \text{front}(u, v), \langle y, x \rangle \rangle \\ y_{\text{shape}} : \langle \lambda v : Ind . \text{cylinder}(v), \langle y \rangle \rangle \end{array} \right] \end{array} \right] . < ttr_{\text{clevr}}.AttrLookupobjectat0x7fa88df06780 >$$

Witness record, s (extracted from scene graph):

	$x=11568_{-7}$		
	$x_{color}=\left\langle \begin{array}{l} \text{color}=\text{yellow} \\ \text{size}=\text{small} \\ \text{rotation}=140.80165619282167 \\ \text{shape}=\text{cube} \\ 3d_{coords}=[1.6490041017532349, -1.4499766826629639, 0.3499999940395355] \\ \text{material}=\text{metal} \\ \text{pixel}_{coords}=[246, 201, 9.080536842346191] \\ \text{id}=11568_{-7} \\ \text{right}=[11568_{-0}, 11568_{-3}, 11568_{-4}] \\ \text{behind}=[11568_{-0}, 11568_{-1}, 11568_{-2}, 11568_{-3}, 11568_{-4}, 11568_{-6}] \\ \text{front}=[11568_{-5}] \\ \text{left}=[11568_{-1}, 11568_{-2}, 11568_{-5}, 11568_{-6}] \end{array} \right\rangle$		
	$x_{size}=\left\langle \begin{array}{l} \text{color}=\text{yellow} \\ \text{size}=\text{small} \\ \text{rotation}=140.80165619282167 \\ \text{shape}=\text{cube} \\ 3d_{coords}=[1.6490041017532349, -1.4499766826629639, 0.3499999940395355] \\ \text{material}=\text{metal} \\ \text{pixel}_{coords}=[246, 201, 9.080536842346191] \\ \text{id}=11568_{-7} \\ \text{right}=[11568_{-0}, 11568_{-3}, 11568_{-4}] \\ \text{behind}=[11568_{-0}, 11568_{-1}, 11568_{-2}, 11568_{-3}, 11568_{-4}, 11568_{-6}] \\ \text{front}=[11568_{-5}] \\ \text{left}=[11568_{-1}, 11568_{-2}, 11568_{-5}, 11568_{-6}] \end{array} \right\rangle$		
	$x_{material}=\left\langle \begin{array}{l} \text{color}=\text{yellow} \\ \text{size}=\text{small} \\ \text{rotation}=140.80165619282167 \\ \text{shape}=\text{cube} \\ 3d_{coords}=[1.6490041017532349, -1.4499766826629639, 0.3499999940395355] \\ \text{material}=\text{metal} \\ \text{pixel}_{coords}=[246, 201, 9.080536842346191] \\ \text{id}=11568_{-7} \\ \text{right}=[11568_{-0}, 11568_{-3}, 11568_{-4}] \\ \text{behind}=[11568_{-0}, 11568_{-1}, 11568_{-2}, 11568_{-3}, 11568_{-4}, 11568_{-6}] \\ \text{front}=[11568_{-5}] \\ \text{left}=[11568_{-1}, 11568_{-2}, 11568_{-5}, 11568_{-6}] \end{array} \right\rangle$		
	$x_{shape}=\left\langle \begin{array}{l} \text{color}=\text{yellow} \\ \text{size}=\text{small} \\ \text{rotation}=140.80165619282167 \\ \text{shape}=\text{cube} \\ 3d_{coords}=[1.6490041017532349, -1.4499766826629639, 0.3499999940395355] \\ \text{material}=\text{metal} \\ \text{pixel}_{coords}=[246, 201, 9.080536842346191] \\ \text{id}=11568_{-7} \\ \text{right}=[11568_{-0}, 11568_{-3}, 11568_{-4}] \\ \text{behind}=[11568_{-0}, 11568_{-1}, 11568_{-2}, 11568_{-3}, 11568_{-4}, 11568_{-6}] \\ \text{front}=[11568_{-5}] \\ \text{left}=[11568_{-1}, 11568_{-2}, 11568_{-5}, 11568_{-6}] \end{array} \right\rangle$		
	$v1x_{right}=\left\langle \begin{array}{l} \text{color}=\text{gray} \\ \text{size}=\text{large} \\ \text{rotation}=187.43182729146503 \\ \text{shape}=\text{cylinder} \\ 3d_{coords}=[-0.5514822602272034, 2.3007681369781494, 0.699999988079071] \\ \text{material}=\text{metal} \\ \text{pixel}_{coords}=[293, 96, 12.618401527404785] \\ \text{id}=11568_{-0} \\ \text{right}=[11568_{-4}] \\ \text{behind}=[11568_{-3}] \\ \text{front}=[11568_{-1}, 11568_{-2}, 11568_{-4}, 11568_{-5}, 11568_{-6}, 11568_{-7}] \\ \text{left}=[11568_{-1}, 11568_{-2}, 11568_{-3}, 11568_{-5}, 11568_{-6}, 11568_{-7}] \end{array} \right\rangle$		
		$\left\langle \begin{array}{l} \text{color}=\text{yellow} \\ \text{size}=\text{small} \\ \text{rotation}=140.80165619282167 \\ \text{shape}=\text{cube} \\ 3d_{coords}=[1.6490041017532349, -1.4499766826629639, 0.3499999940395355] \\ \text{material}=\text{metal} \\ \text{pixel}_{coords}=[246, 201, 9.080536842346191] \\ \text{id}=11568_{-7} \\ \text{right}=[11568_{-0}, 11568_{-3}, 11568_{-4}] \\ \text{behind}=[11568_{-0}, 11568_{-1}, 11568_{-2}, 11568_{-3}, 11568_{-4}, 11568_{-6}] \\ \text{front}=[11568_{-5}] \\ \text{left}=[11568_{-1}, 11568_{-2}, 11568_{-5}, 11568_{-6}] \end{array} \right\rangle$	
		$\left\langle \begin{array}{l} \text{color}=\text{yellow} \\ \text{size}=\text{small} \\ \text{rotation}=140.80165619282167 \\ \text{shape}=\text{cube} \\ 3d_{coords}=[1.6490041017532349, -1.4499766826629639, 0.3499999940395355] \\ \text{material}=\text{metal} \\ \text{pixel}_{coords}=[246, 201, 9.080536842346191] \\ \text{id}=11568_{-7} \\ \text{right}=[11568_{-0}, 11568_{-3}, 11568_{-4}] \\ \text{behind}=[11568_{-0}, 11568_{-1}, 11568_{-2}, 11568_{-3}, 11568_{-4}, 11568_{-6}] \\ \text{front}=[11568_{-5}] \\ \text{left}=[11568_{-1}, 11568_{-2}, 11568_{-5}, 11568_{-6}] \end{array} \right\rangle$	

Answer:

$$q_{rec}.clfr(s) = \text{green}$$